

Cell Tracking and Classification for Fluorescent HeLa Cells

Fnu Zaid Roshan Mayers | PI: Dr. Andrew R. Cohen

Summary

The project's primary focus is on classifying cell division events called 'mitosis' by using segmentation and tracking data given to us from the Cell Tracking Challenge Dataset. The reason for choosing this dataset is because it offers a wide range of raw image sequences along with ground truth labels for cell segmentation and tracking which gives us hands-on experience to deal with real world data which also aligns with our course focus on pattern recognition. By leveraging segmentation masks and ground truths for tracking the cells, we aim to develop a rule-based classifier to distinguish mitotic cells. Preliminary work includes a visualization of cells in frame 1 showing segmented cells with centroids for unique identification to showcase the ability to interact with the Hela Cells Dataset. The impact of this project lies in developing methods to further understand cell behavior and develop unique methods to help biomedical research.

Preliminary Results

In my initial analysis, I was able to visualize frame 1 (t000.tif) of the dataset by applying the segmentation boundaries using *visboundaries* function and the tracking labels with the help of the *regionprops* function in MATLAB. This visualization helped me demonstrate the groundwork of tracking cells across time. Figure 1 highlights the segmentation boundaries (in red) and cell centroids (marked with green dots), with yellow text indicating the tracking IDs. I observed that the 01_GT/SEG directory did not contain all segmentation labels for each frame, while the 01_ST/SEG directory had complete and consistent annotations, indicating a possible manual annotation approach for labeling in the former. This finding guided me to use the 01_ST/SEG directory for segmentation, ensuring more consistent data for further analysis.

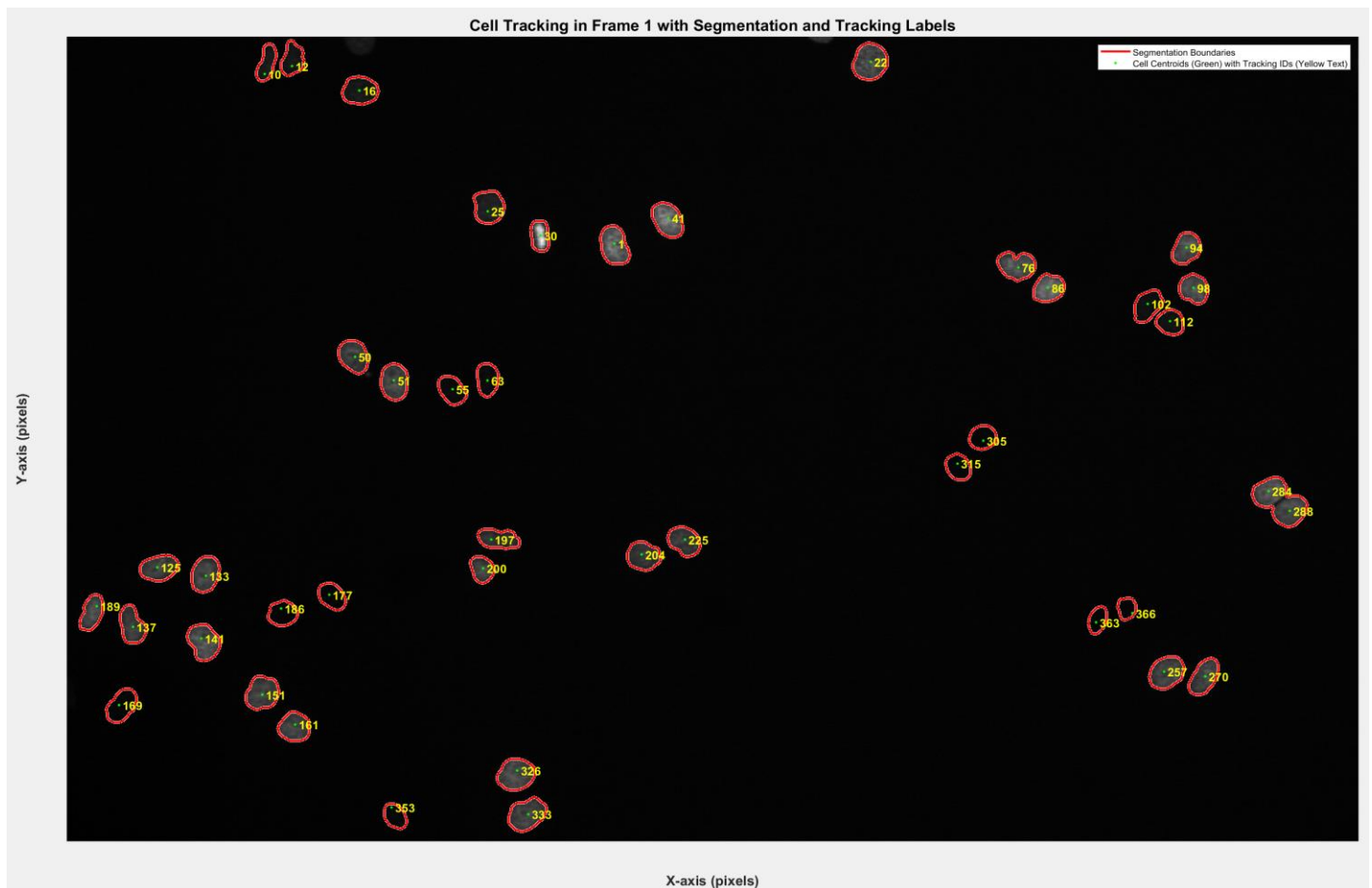


Figure 1: Visualization of Cell Tracking with Segmentation and Tracking Labels of HeLa Cells

Methodology

The approach I will take to detect mitotic events with the dataset centers a rule-based classification system. By analyzing features like cell size, changes in centroid position and tracking number (when the split occurs) shall help me define the rules to identify cells undergoing mitosis. The classifier will leverage the segmentation labels from 01_ST/SEG and tracking labels from the 01_GT/TRA directory, allowing me to be consistent with the detection of mitotic event than relying on probabilistic models. As this project progresses, I may refine these rules based on results and observed data patterns, allowing for adaptations that improve accuracy and responsiveness to changes in cell behavior.

Impact and Innovation

The project addresses a real-world biological challenge and aims to provide an accessible method for cell classification, which is crucial to research, as it aids scientists in tracking and understanding cell behavior in attempts to develop cures to diseases. The project, being rule-based, could potentially prove to be efficient and flexible with changes in patterns and be reliable without the need for complex machine learning models. While advanced deep learning models are often applied, they require significant computational resources and financial investment. The innovation of this project lies in its simplicity and ability to implement and apply the classifier to similar datasets without being heavy on consumer contributions.

Bibliography

Dr. Andrew R. Cohen, a prominent researcher in biomedical imaging and cell tracking, has significantly contributed to methods for particle tracking and image-based analysis of biological systems. His work includes foundational research on tracking methodologies and their applications in cellular studies, as evidenced in his highly cited publications. Dr. Cohen's expertise in computational microscopy and automated cell tracking aligns directly with the goals of this project, making him an ideal PI for the Pattern Recognition course project.

The following works by Dr. Cohen serve as references for this project:

- Maška M, Ulman V, Delgado-Rodriguez P, Gómez-de-Mariscal E, Nečasová T, Guerrero Peña FA, Ren TI, Meyerowitz EM, Scherr T, Löffler K, Mikut R, Guo T, Wang Y, Allebach JP, Bao R, Al-Shakarji NM, Rahmon G, Toubal IE, Palaniappan K, Lux F, Matula P, Sugawara K, Magnusson KEG, Aho L, Cohen AR, Arbelle A, Ben-Haim T, Raviv TR, Isensee F, Jäger PF, Maier-Hein KH, Zhu Y, Ederra C, Urbiola A, Meijering E, Cunha A, Muñoz-Barrutia A, Kozubek M, Ortiz-de-Solórzano C. The Cell Tracking Challenge: 10 years of objective benchmarking. *Nat Methods*. 2023 Jul;20(7):1010-1020. doi: 10.1038/s41592-023-01879-y. Epub 2023 May 18. PMID: 37202537; PMCID: PMC10333123.
- Chenouard N, Smal I, de Chaumont F, Maška M, Sbalzarini IF, Gong Y, Cardinale J, Carthel C, Coraluppi S, Winter M, Cohen AR, Godinez WJ, Rohr K, Kalaidzidis Y, Liang L, Duncan J, Shen H, Xu Y, Magnusson KE, Jaldén J, Blau HM, Paul-Gilloteaux P, Roudot P, Kervrann C, Waharte F, Tinevez JY, Shorte SL, Willemse J, Celler K, van Wezel GP, Dan HW, Tsai YS, Ortiz de Solórzano C, Olivo-Marin JC, Meijering E. Objective comparison of particle tracking methods. *Nat Methods*. 2014 Mar;11(3):281-9. doi: 10.1038/nmeth.2808. Epub 2014 Jan 19. PMID: 24441936; PMCID: PMC4131736.
- Cohen AR, Roysam B, Turner JN. Automated tracing and volume measurements of neurons from 3-D confocal fluorescence microscopy data. *J Microsc*. 1994 Feb;173(Pt 2):103-14. doi: 10.1111/j.1365-2818.1994.tb03433.x. PMID: 8169949.